



SESSION 44

Confidence Intervals

Point vs Interval Estimation · Z-Intervals · T-Intervals · Margin of Error

Point vs Interval Estimation

Confidence Interval Definition

Confidence Interval for Mean (Sigma Known)

Confidence Interval for Mean (Sigma Unknown)

The Student's t-Distribution

Confidence Intervals for Proportions

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Session Notes — Complete Reference

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01 • Point vs Interval Estimation

Point Estimation: A single numerical value used to estimate a population parameter (e.g., sample mean $\bar{X}$ estimating μ). Point estimators are rarely exactly correct.

Interval Estimation: A range of values within which the population parameter is expected to fall, with a specified degree of confidence.

02 · Confidence Interval Definition

A **Confidence Interval (CI)** is an interval computed from sample data that contains the true population parameter with a probability equal to the **Confidence Level** ($1 - \alpha$).

A 95% confidence level means that if we repeat the sampling process 100 times, 95 of the computed intervals will contain the true population parameter.

03 · Confidence Interval for Mean (Sigma Known)

When the population standard deviation σ is known, we use the Standard Normal (Z) distribution:

Z-INTERVAL FORMULA

$$CI = \bar{X} \pm Z_{(\alpha/2)} \cdot (\sigma / \sqrt{n})$$

For a 95% confidence interval, $\alpha = 0.05$ and $Z_{\{\alpha/2\}} = 1.96$.

04 · Confidence Interval for Mean (Sigma Unknown)

In practice, σ is almost never known. We use the sample standard deviation s and the **t-distribution**:

T-INTERVAL FORMULA

$$CI = \bar{X} \pm t_{(\alpha/2, df)} \cdot (s / \sqrt{n})$$

where $df = n - 1$ represents the degrees of freedom.

05 • The Student's t-Distribution

The **t-distribution** is symmetric and bell-shaped but has heavier tails than the standard normal distribution. This accounts for the extra uncertainty of estimating σ with the sample statistic s .

As the degrees of freedom ($df = n - 1$) increase, the t-distribution approaches the standard normal distribution.

06 · Confidence Intervals for Proportions

For categorical variables (e.g., proportion of users clicking a link):

PROPORTION CI FORMULA

$$CI = \hat{p} \pm Z_{(\alpha/2)} \cdot \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

07 · Factors Influencing Margin of Error

The **Margin of Error (MOE)** is the width of half of the confidence interval:

- **Sample size (n):** Increasing n decreases the standard error, narrowing the interval (decreases MOE).
- **Confidence level:** Higher confidence levels (e.g., 99% vs 95%) increase the critical value (Z or t), widening the interval (increases MOE).
- **Data variance:** Higher standard deviation increases MOE.

08 · Python Implementation of Confidence Intervals

```
import numpy as np
from scipy import stats

data = [45, 47, 50, 52, 54, 58, 60, 62, 65, 68]
n = len(data)
mean = np.mean(data)
sem = stats.sem(data) # Standard error of mean: s / sqrt(n)

# 95% Confidence Interval using t-distribution
ci_t = stats.t.interval(confidence=0.95, df=n-1, loc=mean, scale=sem)
print(f"95% CI (t-distribution): {ci_t[0]:.2f} to {ci_t[1]:.2f}")
```

09 · Quick Reference Card



Confidence Intervals Cheat Sheet

- **Sigma Known:** Use Z -interval with critical values (1.96 for 95%).
- **Sigma Unknown:** Use t -interval with $df = n - 1$.
- **Margin of Error:** $\text{MOE} = \text{Critical Value} \times \text{Standard Error}$.
- **Width of CI:** $2 \times \text{MOE}$.